DOCVOICE AI | System Architecture Blueprint

Autonomous 5-Agent Document Intelligence • Dynamic Length Control • Neural Audio Synthesis

FastAPI Async | React 18 (3D) | Edge-TTS Neural | GovReport Eval

1. INGESTION & CONTEXT

[INPUT] Document Ingestion

- Multi-format: PDF, DOCX, TXT
- Magic bytes verification & 10MB limit
- In-memory stream parsing (PyPDF, docx)
- Rejection of empty or corrupted files

[QUERY] Natural Language Focus

- Custom query & analytical focus area
- Word budget (e.g. 'summarize in 100 words')
- Token budget (e.g. 'in 60 tokens')
- Dynamic regex parser extraction

[CHUNK] Sentence Chunker

- Noise scrubbing & whitespace cleanup
- Overlapping sentence chunker (1000 tok)
- Sliding window context preservation
- Indexed with document ID & chunk index

[POOLER] Adaptive Context Pooler

- Scales chunks dynamically to word budget
- Short (<60w): Intro + Conclusion chunks
- Medium (~100w): Multi-chunk balanced sweep
- Extended (200w+): Full corpus coverage

Clean Chunks

2. AUTONOMOUS 5-AGENT CORE & VALIDATION GATEKEEPER

[AGENT 1] Document Analysis

- Deep semantic & structural inspection
- Identifies core themes, topics & tone
- Discovers numerical & financial claims
- Formats structured AnalysisResult
- Emits: AnalysisResult contract

Themes & Context

[AGENT 2] Summarization Engine

- Dynamic Word/Token Budget Control
- Strict Single-Paragraph Mandate
- Heuristic sentence extraction & Gemini
- Adaptive length scaling without loss
- Emits: SummaryDraft contract

Draft Summary

[AGENT 3] Summary Validation

- Verifies numerical facts against chunks
- Calculates source chunk token overlap
- Detects & scrubs hallucinated claims
- Enforces length ceiling & paragraph flow
- Emits: ValidationReport contract

Audit Pass

[GATEKEEPER] TTS Safety Barrier

- Strict Hard Barrier: is_approved_for_tts
- HTTP 403 Forbidden lock on rejected drafts
- Trims length cleanly at clause boundaries
- Blocks speech engine from ungrounded text
- Emits: ValidatedSummaryPayload

Approved Summary

[AGENT 4] Quality Evaluation

- Multi-dimensional scoring (0-1.0):
  * Factual accuracy & Groundedness
  * Coherence & Readability index
- GovReport academic benchmark telemetry
- Automated ROUGE-1, ROUGE-2, ROUGE-L
- Emits: QualityScore contract

ROUGE Metrics

TTS Payload

[AGENT 5] Neural Voice Gen

- Microsoft Edge-TTS Neural Synthesizer
- Multilingual voice matrix (8 voices):
  * English (Andrew / Aria Neural)
  * Kannada (Gagan / Sapna Neural)
  * Hindi (Madhur / Swara Neural)
  * Telugu (Mohan / Shruti Neural)
- Outputs: High-fidelity streaming MP3

Streamed MP3

3. PRESENTATION & BENCHMARK

[UI] React 18 + Three.js 3D

- Interactive 3D particle sphere visualizer
- Real-time audio frequency waveform player
- Word count & token telemetry meters
- Single-paragraph clean typography

[VOICE] Hands-Free Voice Mode

- Web Speech API microphone input
- Voice-prompted questions & queries
- Auto-play spoken audio responses
- Full visual-impairment accessibility

[EVAL] GovReport Benchmark

- Live streaming GAO / CRS federal reports
- Automated ROUGE-1 / 2 / L computation
- Hallucination rate & groundedness logs
- Academic gold-standard verification

[API] FastAPI & SQLite Storage

- Sub-millisecond async route dispatch
- SQLite ACID document & chunk repository
- Complete A2A message transaction log
- Auto-generated OpenAPI / Swagger docs

FastAPI Async | React 18 (3D) | 5 Autonomous Agents | Word/Token Budget Engine | Strict Safety Barrier | Edge-TTS Neural (4 Langs) | GovReport Benchmark | 29/29 Passing Pytest